Farmora Investor & Academic Pitch Deck

Connecting Soil to Shelf ? Fair, Fresh, and Fast

SLIIT Department of Software Engineering | Milestone 5 & Final Evaluation

Slide 1: Title Slide

- Title:** Farmora
- Subtitle:** The Next-Generation Smart Agricultural Marketplace & Traceable Cold-Chain Ecosystem
- Presented by:** Team Farmora (Ahsan Mohammed, Kajana S., Sureka)
- Institution:** Sri Lanka Institute of Information Technology (SLIIT)

Slide 2: The Agricultural Crisis in Sri Lanka

- The Problem:**
- 40% Loss to Middlemen:** Fragmented broker cartels at physical economic centres exploit smallholder farmers.
- 30% Post-Harvest Spoilage:** Lack of verified cold-chain logistics causes fresh perishables to rot during transit.
- Zero Traceability:** Supermarkets and hospitality buyers cannot verify chemical safety, harvest date, or origin.
- Payment Defaults:** High risk of bad debts and delayed cash disbursements.

Slide 3: The Solution ? Farmora Ecosystem

- Direct Digital Marketplace:** Farmers sell directly to supermarkets, hotel syndicates, and bulk buyers.
- Field Video Verification:** Farmers upload 10-second proof-of-freshness video directly from the field.
- Cryptographic Batch QR:** Unique QR codes attached to crates verify harvest date, farmer identity, and agronomy data.
- Guaranteed Escrow Protection:** Funds locked at order placement; released within 2 seconds of verified delivery.
- Integrated Cold-Chain Fleet Hub:** Real-time IoT temperature monitoring across all transport reefers.

Slide 4: Multi-Role Experience

- For Farmers:** 1-tap listing, voice search, market price ticker, instant escrow payouts.
- For Wholesale Buyers:** Transparent grades, field video inspection, bulk procurement, dispute resolution.
- For Fleet Transporters:** Route matching, vehicle & driver depot management, live GPS, digital POD.
- For Admin / Regulators:** Escrow treasury settlement, user KYC validation, dispute resolution tribunal.

Slide 5: Technical Excellence & Architecture

- Front-End:** Flutter 3.x (Multi-platform compilation to Android Release APK & Progressive Web App).
- Back-End:** Firebase Serverless (Cloud Firestore real-time streams, Cloud Storage, Authentication).
- Media Engine:** Cloudinary API for optimized video streaming and automatic cleanup upon order fulfillment.
- Security:** Granular Role-Based Access Control (RBAC) enforced at Firestore engine level.
- Offline First:** Local caching enables uninterrupted harvest logging in low-connectivity rural farms.

Slide 6: Usability & Testing Validation (Milestone 3)

- Usability Benchmark:** System Usability Scale (SUS) Score of **84.5 / 100 (Grade A)**.
- Task Success Rate:** Over 95% across critical user paths.
- Accessibility:** High-contrast daylight UI (WCAG 2.1 AA) with full trilingual support (Sinhala, Tamil, English).

Slide 7: Business Model & Revenue Streams

1. **Marketplace Commission:** 2.5% take rate on wholesale transactions.
2. **Escrow Guarantee Fee:** 1.0% platform assurance fee.
3. **Logistics Dispatch Matching:** 3.5% freight matching commission.
4. **Enterprise SaaS:** $25/month for supermarket inventory API synchronization.

- Total Addressable Market (TAM):** $420M+ agricultural trade volume in Sri Lanka.

Slide 8: Go-to-Market & Roadmap

- Q1:** Pilot rollout in Dambulla & Nuwara Eliya agricultural belts (250 verified farmers).
- Q2:** Formal procurement integration with supermarket chains (Keells, Cargills).
- Q3:** Full logistics fleet matching rollout across North & Central Provinces.
- Q4:** Regional expansion and agri-microcredit underwriting partnerships.

Slide 9: The Deliverables

- Live Progressive Web App:** https://farmingapp-24b34.web.app
- Production Android APK:** ‘build/app/outputs/flutter-apk/app-release.apk‘ (99.3MB, compileSdk 36)
- Open GitHub Repository:** https://github.com/AHSANMOHAMMED/Farmora-App.git
- Complete Project Report:** Milestones 1 through 6 documentation.

Slide 10: Conclusion & Q&A

- "Farmora connects soil to shelf ? ensuring fair profits for farmers, guaranteed freshness for buyers, and zero wasted food."*
- Thank you! We welcome your questions.**